SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

FOR

SECURITY INCIDENT MAPPING SYSTEM (SIM)

A Web-Based Security Incident Reporting and Mapping Platform

Submitted By

Ogunbodede ahmed toluwalase

241201175

Computer science

Department: Computer Science

Course Code: SEN 202

Course Title: Introduction to Software Engineering 2

Institution: Adekunle Ajasin University, Akungba-Akoko (AAUA)

Date: July 2026

TABLE OF CONTENTS

1. INTRODUCTION

1.1 Background

Security challenges have become a major concern in many communities, including university campuses. Incidents such as theft, assault, harassment, and other criminal activities often occur due to poor communication channels between community members and security authorities.

The Security Incident Mapping (SIM) system is a proposed web-based platform designed to improve security reporting and monitoring. The system allows students, staff, and residents within the university community to report security incidents digitally while enabling security personnel to monitor, manage, and analyze reported cases.

By providing a centralized reporting platform, SIM aims to improve response time, increase security awareness, and support better decision-making through organized crime data.

2. PURPOSE OF THE SYSTEM

The purpose of the Security Incident Mapping System is to provide an efficient platform where users can quickly report security incidents and where authorized security personnel can manage and analyze these reports.

The system is designed to:

Provide an easier method of reporting security issues.

Reduce delays associated with manual reporting methods.

Allow security personnel to monitor incidents effectively.

Provide location-based information through mapping features.

Improve safety awareness within the community.

3. SCOPE OF THE SYSTEM

The Security Incident Mapping System will provide the following services:

User registration and authentication.

Online reporting of security incidents.

Categorization of reported incidents.

Location mapping of reported cases.

Incident status tracking.

Security updates and notifications.

Administrative management of incident reports.

The system will focus on improving security communication between community members and security authorities.

4. SYSTEM OVERVIEW

4.1 Product Perspective

SIM is a web-based application that serves as a communication bridge between members of the community and security personnel. It replaces traditional reporting methods by providing a digital platform where security information can be collected, stored, and managed.

The system will contain:

User interface for reporting incidents.

Database for storing incident information.

Mapping component for displaying incident locations.

Administrative dashboard for security officers.

5. USER CLASSES AND CHARACTERISTICS

5.1 Students/Residents

Users in this category can:

Create an account.

Log into the system.

Submit security reports.

View security announcements.

Track submitted reports.

5.2 Security Personnel/Admin

Administrators can:

Access submitted reports.

Verify incident information.

Update incident status.

Remove invalid reports.

Generate security reports.

6. FUNCTIONAL REQUIREMENTS

6.1 User Authentication

The system shall allow users to create accounts and securely log in using their credentials.

6.2 Incident Reporting

The system shall allow users to submit security incidents containing:

Incident category.

Description of the event.

Date and time.

Location.

Supporting evidence where necessary.

6.3 Incident Mapping

The system shall display reported incidents on a digital map to help identify areas with frequent security issues.

6.4 Incident Management

The administrator shall be able to:

View reported incidents.

Verify reports.

Update incident progress.

Manage stored records.

6.5 Notification System

The system shall provide users with security alerts and updates concerning incidents.

6.6 Search and Filtering

The system shall allow users and administrators to search incidents based on:

Location.

Incident type.

Date reported.

Status.

7. NON-FUNCTIONAL REQUIREMENTS

7.1 Security

User information must be protected.

Unauthorized users must not access administrative features.

Data transmission should be secure.

7.2 Performance

The system should respond quickly to user requests.

It should support multiple users accessing the platform.

7.3 Reliability

The system should maintain accurate and consistent records.

Data should not be lost during normal operation.

7.4 Usability

The interface should be simple and easy to understand.

Users should be able to report incidents without difficulty.

7.5 Availability

The system should be available whenever users need to access security services.

8. EXTERNAL INTERFACE REQUIREMENTS

8.1 User Interface

The system will include:

Login page.

Registration page.

User dashboard.

Incident reporting form.

Security map interface.

Administrator dashboard.

8.2 Hardware Interface

The system requires:

Smartphone, tablet, or computer.

Internet connection.

8.3 Software Interface

The system interacts with:

Web browsers.

Database management systems.

Mapping services.

9. SYSTEM FEATURES

9.1 Incident Reporting Module

Allows users to submit details about security incidents.

9.2 Mapping Module

Displays incident locations visually for better monitoring.

9.3 Administration Module

Allows authorized personnel to control and manage reports.

9.4 Notification Module

Provides security updates and important announcements.

10. DATABASE REQUIREMENTS

The system database will contain the following entities:

User Table

Attributes:

User_ID

Full Name

Email

Password

User Type

Incident Table

Attributes:

Incident_ID

User_ID

Incident Category

Description

Location

Date

Status

Administrator Table

Attributes:

Admin_ID

Name

Username

Password

11. SYSTEM CONSTRAINTS

The system may be affected by:

Availability of internet connection.

Accuracy of information provided by users.

Availability of location services.

Security and privacy requirements.

12. FUTURE ENHANCEMENTS

Future improvements may include:

Development of a mobile application version.

Artificial Intelligence-based crime prediction.

Emergency panic button integration.

Connection with official campus security systems.

Real-time incident tracking.

13. CONCLUSION

The Security Incident Mapping System provides an effective solution for improving security reporting and monitoring within a community. By allowing users to report incidents digitally and enabling security personnel to analyze reports efficiently, the system can contribute to faster response times and improved safety.

The implementation of SIM will create a more organized and reliable approach to handling security challenges through technology.